

CAMBRIDGE INTERNATIONAL AS LEVEL BIOLOGY (9700)

TOPIC 1 — CELL STRUCTURE

Microscopy, Organelles & Cell Ultrastructure

CANDIDATE

Hamna (Grade A* Scholar)

SPECIFICATION

Cambridge 9700 AS Level

TOTAL MARKS / QUESTIONS

6 Marks · 1 Questions

EXAMINATION STRUCTURE & 80/20 QUESTION TARIFF

- Section A — Extended High-Tariff Questions: 20 × 6 Marks (120 Marks total) [Multi-part synoptic evaluation]
- Section B — Data Interpretation & Structured Explanations: 20 × 4 Marks (80 Marks total)
- Section C — Core Definitions & Quantitative Calculations: 10 × 2 Marks (20 Marks total)
- Section D — 10 High-Frequency Core Repeated Questions for Exam Mastery & Knowledge Consolidation
- Division: 80% Authentic Cambridge 9700 Past Paper references + 20% Original A* Extension Problems

SYLLABUS SUBTOPICS COVERED IN THIS PACK

- 1.1 Microscopy
- 1.2 Organelles

EXAMINER'S DIRECTIVE FOR CANDIDATE HAMNA

Write legibly in the dotted spaces provided. Pay strict attention to Cambridge command words (Explain, Describe, Deduce, Calculate).
Where biological diagrams or graphs are provided, extract exact qualitative and quantitative coordinates to support your answers.

SECTION A — EXTENDED STRUCTURED QUESTIONS

1 9700/22/M/J/23/Q1 Eukaryotic Organelles

[6 Marks] Syllabus 1.3

Fig. 1.1 shows a transmission electron micrograph (TEM) diagram of an animal cell.

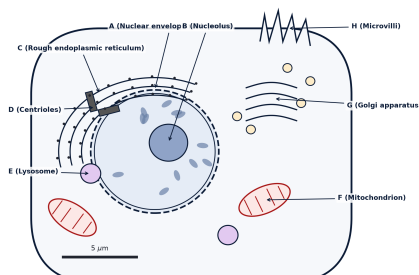


Fig. 1.1: Transmission electron micrograph (TEM) diagram of an animal cell

Fig. 1.1: Transmission electron micrograph (TEM) diagram of an animal cell showing ultrastructure.

(a) Identify the structures labelled A, B, and C in Fig. 1.1.

[3]

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(b) Describe the structural role of organelle F in aerobic respiration.

[3]

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EXAMINER MARK SCHEME & MARKING GUIDELINES

TOPIC 1 — CELL STRUCTURE — Cambridge International AS Level Biology (9700)

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
1	(a)	A: Nuclear envelope / double nuclear membrane [1]; B: Nucleolus [1]; C: Rough endoplasmic reticulum / RER [1]	[3]
1	(b)	Inner mitochondrial membrane is folded into cristae to provide large surface area for ATP synthase / electron transport chain enzymes [1]; mitochondrial matrix contains 70S ribosomes and circular DNA for synthesizing mitochondrial proteins [1]; site of Krebs cycle and oxidative phosphorylation [1]	[3]